

Automation in Marine Industry

Manufacturing Concepts, Shipbuilding Automation, Digital Shipyards & Robotics

Amit Kumar Singh

Assistant Professor (ECE)

School of Naval Architecture and Ocean Engineering

Indian Maritime University Visakhapatnam Campus, Sabbavaram Mandal

Visakhapatnam, Andhra Pradesh – 531 035

amitkumarsingh@imu.ac.in

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- 1 Concepts in Manufacturing and Automation
- 2 Reasons for Automating
- 3 Types of Production
- 4 Types of Automation
- 5 Automation Strategies
- 6 Levels of Automation in Shipbuilding
- 7 Digital Shipyards
- 8 Robots in Welding
- 9 Robots in Blasting and Painting
- 10 Heavy Lifting and Handling
- 11 Automation of Other Tasks

What is Manufacturing?

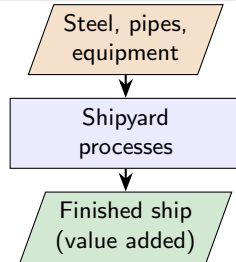
Two views of manufacturing

Technological: applying physical and chemical processes to change the geometry, properties or appearance of a material to make parts or products.

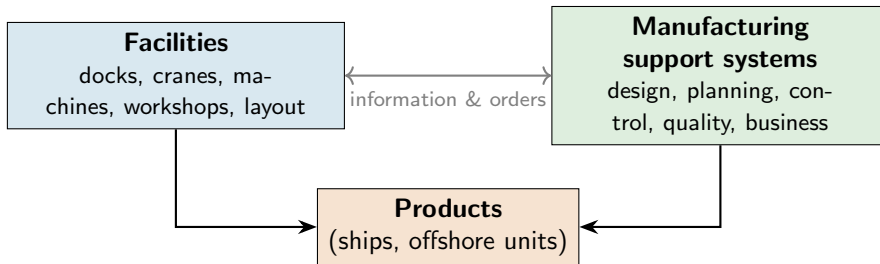
Economic: transforming materials into items of **greater value** by processing and assembly — *value is added*.

- Four basic **manufacturing operations**:

- ① **Processing** — cutting, bending, welding, painting.
- ② **Assembly** — joining parts into blocks, blocks into a hull.
- ③ **Material handling** — cranes, transporters, conveyors.
- ④ **Inspection & test** — NDT, dimensional checks, sea trials.



The Production System



Automation of facilities

CNC cutting, welding robots, automatic panel lines, crane control — **automated manufacturing systems**.

Computerised support systems

CAD/CAM, production planning, ERP, MES — **computer-integrated manufacturing (CIM)**.

Production Metrics: Worked Examples

Key definitions

Cycle time T_c — time per part at a station.

Production rate $R_p = 60/T_c$ (parts/h).

Availability $A = \frac{\text{MTBF} - \text{MTTR}}{\text{MTBF}}$

Manufacturing lead time

$$\text{MLT} = n_o \left(\frac{T_{su}}{Q} + T_c + T_{no} \right)$$

n_o = operations, T_{su} = set-up time, Q = batch size,

T_{no} = non-operation (waiting, moving) time.

1. A stiffener-cutting station has $T_c = 6$ min:

$$R_p = 60/6 = \boxed{10 \text{ parts/h}}$$

2. A plasma cutter: MTBF = 50 h, MTTR = 5 h:

$$A = (50 - 5)/50 = \boxed{0.90}$$

3. A batch of $Q = 20$ panels passes $n_o = 4$ operations, each with $T_{su} = 3$ h, $T_c = 0.5$ h, $T_{no} = 6$ h:

$$\text{MLT} = 4(0.15 + 0.5 + 6) = \boxed{26.6 \text{ h}}$$

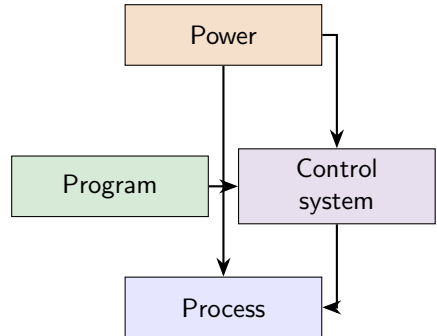
Waiting time dominates — automation of *handling* and *planning* often saves more than faster machines.

What is Automation?

Automation

Technology by which a process or procedure is performed with **minimum human assistance**, using **power**, a **program of instructions** and a **control system**.

- **Mechanisation** — machines replace human *muscle* (a powered welding carriage).
- **Automation** — machines also replace human *control and decisions* (a robot that finds and welds the seam itself).
- **Computerisation** — information flow is automated (CAD model → robot program).



Reasons for Automating

General reasons

- Increase labour **productivity**.
- Reduce labour cost; overcome labour **shortages**.
- Reduce or remove routine manual tasks.
- Improve worker **safety**.
- Improve product **quality**.
- Reduce manufacturing **lead time**.
- Do what cannot be done manually.
- Avoid the high cost of *not* automating.

Especially in shipbuilding

- Shortage of **skilled welders** and fitters.
- Welding in **confined** double-bottom cells and tanks — heat, fumes.
- Blasting & painting — dust, solvents, health hazards.
- Tight delivery schedules and global price competition.
- Class & coating rules demand documented, consistent quality.

The Case For and Against Automation

For	Against / cautions
Higher output per worker and per shift	High capital cost; long payback for low volumes
Consistent weld and paint quality	Every ship is different — programming effort
Safer, healthier working conditions	Large, heavy, curved parts are hard to fixture
Better data for planning and costing	Needs skilled technicians & maintenance
Competitiveness against low-wage yards	Workforce concerns — retraining required

The USA principle

Understand the existing process → **S**implify it → **A**utomate it. In shipbuilding, simplification (standard details, straight panels, fewer part types) is often the biggest step.

Types of Production

Low quantity (job shop)

1–100 units / year

- Special, customised products.
- General-purpose equipment.
- **Ships**, offshore units.

Medium quantity (batch)

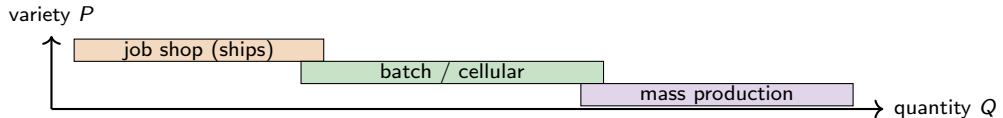
100–10 000 / year

- Batches, then change-over.
- Cells for part families.
- Engines, pumps, boats.

High quantity (mass)

> 10 000 / year

- Flow-line production.
- Special-purpose equipment.
- Fasteners, cars.



Where Does Shipbuilding Fit?

A paradox

The **ship** is a one-off (job-shop) product, but it is made from **thousands of similar parts**: plates, stiffeners, brackets, pipe spools, panels.

- **Group technology**: group similar interim products into families (flat panels, curved blocks, pipe spools).
- Each family is built on a dedicated **work station or line** — batch-like and automatable.
- Result: a *job-shop product* built with *flow-line methods*.

Item in one ship	Approx. count
Steel parts	20 000–50 000
Flat panels	100–300
Hull blocks	50–200
Pipe spools	5 000–15 000
Weld length	100s of km

Indicative figures for a large merchant ship.

Plant Layouts in a Shipyard

Layout	Idea	Shipyard example
Fixed-position	product stays; workers & equipment come to it	hull erection in the building dock
Process	machines grouped by process type	plate shop, pipe shop, paint shop
Cellular	cells dedicated to part families	sub-assembly cells, pipe-spool cells
Product (flow line)	stations in sequence; work flows through	automatic panel line

Key idea

Modern yards move as much work as possible **upstream** — from the dock (fixed-position, hard to automate) into workshops with cellular and flow-line layouts (easy to automate).

Types of Automation

Fixed

Hard automation

- Sequence fixed by equipment design.
- High volume, one product.
- **Yard:** shot-blasting & priming line for plates, profile cutting line.

Programmable

Change the program

- New part = new program.
- Batch production.
- **Yard:** CNC plasma cutting, robot welding of sub-assemblies.

Flexible

No change-over loss

- Mix of different parts.
- Programs made off-line.
- **Yard:** panel line with robots programmed automatically from CAD.

Which fits shipbuilding?

Because every ship is different, shipyards rely mainly on **programmable and flexible** automation — and on getting the robot programs **directly from the 3-D product model**.

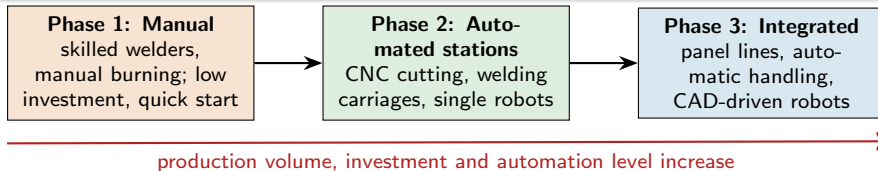
Ten Strategies for Automation and Process Improvement

#	Strategy	Shipyard example
1	Specialisation of operations	dedicated stiffener-welding gantry
2	Combined operations	cut, mark and bevel a plate in one CNC set-up
3	Simultaneous operations	several welding heads on one panel at once
4	Integration of operations	plate line: blasting → priming → cutting linked by conveyors
5	Increased flexibility	robots programmed per panel from CAD
6	Improved material handling	block transporters, automated plate stock-yard
7	On-line inspection	laser scanning of block dimensions (accuracy control)
8	Process control & optimisation	adaptive welding with seam tracking
9	Plant operations control	MES tracking every block and pipe spool
10	Computer-integrated manufacturing	one product model from design to delivery (<i>digital ship-yard</i>)

Automation Migration Strategy

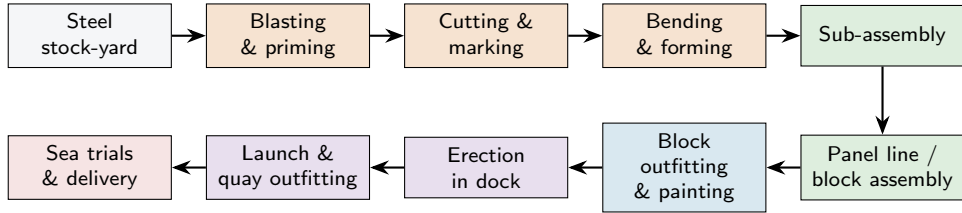
Step-by-step path

New products or processes are rarely automated in one step. The **migration strategy** moves from manual to fully integrated production as demand and experience grow.



- Reduces risk — each step is proven before the next investment.
- Uses the same product model data at every phase so programs and experience carry forward.

The Shipbuilding Production Process



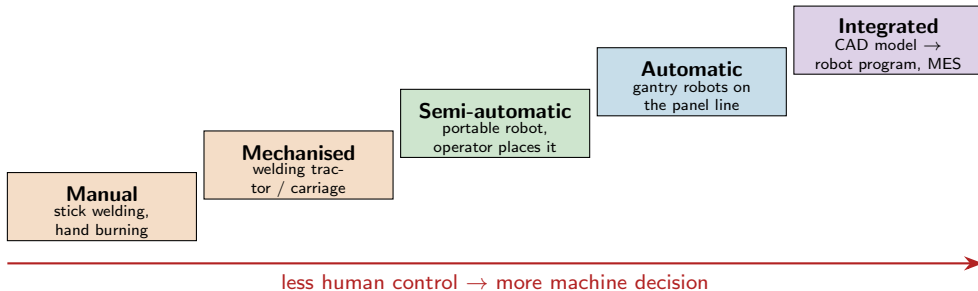
Hull block construction (HBCM)

The hull is divided into **blocks** (50–900 t) built in parallel in workshops, then **erected** in the dock.

Zone outfitting & painting

Pipes, cables and equipment are fitted and blocks are painted **before** erection — easier access, less work in the dock.

Levels of Automation (Degree of Automation)



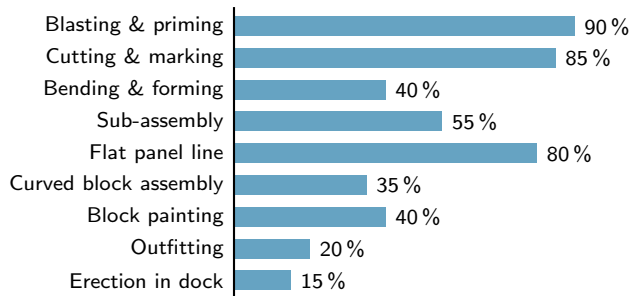
Hierarchy levels

Device (sensor, motor) → **machine** (robot, CNC) → **cell / line** (panel line) → **plant** (yard MES) → **enterprise** (ERP, supply chain).

Rule of thumb

The more **repetitive** and **flat / straight** the work, the higher the level of automation that pays off.

Level of Automation by Production Stage



Indicative degree of automation in a modern yard.

- **Highest** where parts are flat, straight and repeated: plate lines, cutting, flat panels.
- **Lowest** where work is 3-D, confined or unique: curved blocks, outfitting, erection.
- Research focus: mobile robots for **confined spaces** and **erection joints**.

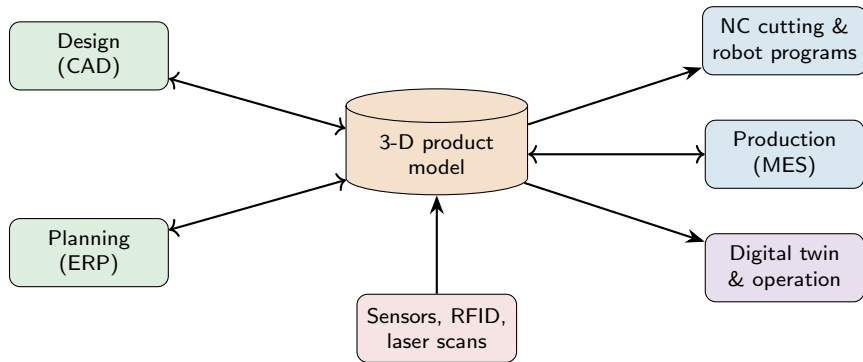
What is a Digital Shipyard?

Digital shipyard (Shipyard 4.0)

A shipyard where design, planning, production and after-sales share **one digital product model** and **real-time data**, connected by networks, sensors and software — Industry 4.0 applied to shipbuilding.

- **3-D product model** — single source for drawings, part lists, NC and robot programs.
- **Digital twin** — virtual copy of the ship and of the yard, updated with real data.
- **IoT & RFID** — track plates, blocks, pipe spools, tools and people.
- **Simulation** — test the production plan before cutting steel.
- **MES / ERP** — plan and follow every work package.
- **AR / VR** — workers see the model overlaid on the real block.
- **3-D laser scanning** — compare built blocks with the model (accuracy control).
- **Data analytics & AI** — predict delays, optimise crane use.

The Digital Thread in a Shipyard



- Data is created **once** and reused everywhere — no re-drawing, no re-typing.
- Feedback from the shop floor (as-built scans, progress) flows **back** to planning.
- The same twin supports the ship's operation, maintenance and repair after delivery.

Benefits and Challenges of Digital Shipyards

Benefits

- Shorter design-to-production time.
- Fewer errors, less rework on block fit-up.
- Real-time visibility of progress & costs.
- Automatic robot & NC programming.
- Better planning of cranes, docks, workforce.
- Lifetime data for owners (maintenance, retrofit).

Challenges

- High IT investment and integration effort.
- Legacy systems and data formats.
- Harsh environment for sensors & networks (steel, dust, water).
- Cyber-security.
- Training & change of work culture.

Welding: the Biggest Job in a Shipyard

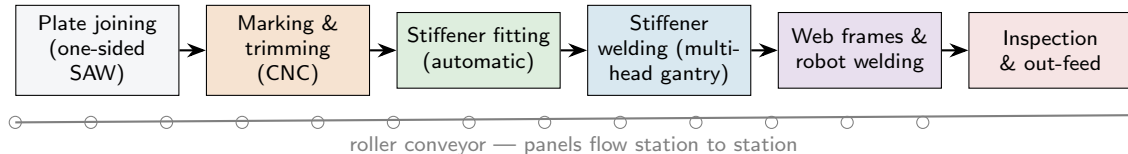
- Welding is roughly **a third or more** of hull steel-work labour.
- Processes: SAW (submerged arc), FCAW / MIG-MAG, one-sided welding with ceramic backing, laser-hybrid welding.
- Joints: long butt welds of plates, fillet welds of stiffeners and webs, short welds in confined cells.

Arc-on time

A manual welder spends much time on positioning, cleaning and moving — the arc is on only **20–35 %** of the time. A robot can reach **70–90 %**.

Robot type	Use in the yard
Welding tractor / carriage	long straight butt & fillet welds
Gantry robots	panel line: stiffeners, webs, collars
Portable robots	placed by crane into double-bottom cells
Mobile / crawler	vertical & erection joints on the hull
Pipe-welding cells	orbital welding of pipe spools

Automatic Panel Line



- A **flow line** for flat panels (e.g. 20 m × 20 m).
- Panels move; equipment stays — the product layout.
- Gantry robots get their weld paths **from the CAD model**.

Typical results

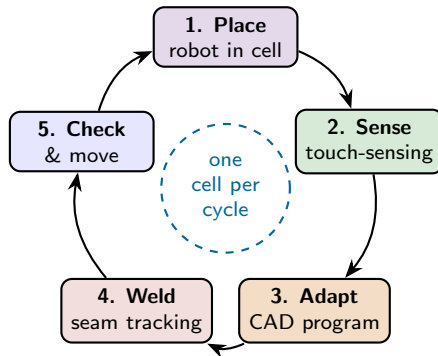
Several times the productivity of manual assembly, far less distortion (controlled heat input), and consistent weld quality.

How a Portable Welding Robot Works: The Cell Cycle

The Place—Sense—Weld—Move cycle

Like the CPU's machine cycle, a portable robot repeats one cycle for every **double-bottom cell** (a box formed by floors and girders) until the block is finished.

- 1 **Place** — crane or operator puts the robot into the next cell.
- 2 **Sense** — touch-sensing with the wire finds the real plate edges.
- 3 **Adapt** — program from CAD is corrected to the measured geometry.
- 4 **Weld** — all fillet welds in the cell, with arc seam tracking.
- 5 **Check & move** — report welds done, lift robot to the next cell.

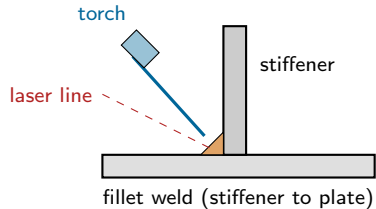


Off-line programming (OLP)

Weld paths, torch angles and parameters are generated **automatically from the 3-D model** — essential because every panel is different. Teaching by pendant would take longer than welding.

Seam finding & tracking

Touch sensing (wire touches the plate), **arc sensing** (current changes during weaving), **laser line scanners** — correct for fit-up gaps and distortion.



- **Multi-robot gantries** weld both sides of a stiffener at once — less distortion.
- **Weld data logging** gives traceable quality records for class societies.

Pseudocode

- 1 START
- 2 Set speeds & arc-on factors: manual 0.3 m/min, 30%; robot 0.5 m/min, 80%
- 3 READ weld length L
- 4 $t = L / (v * f) / 60$ for each method (hours)
- 5 PRINT both times
- 6 PRINT % time saved
- 7 STOP

C Program (welding time)

```
#include <stdio.h>

int main(void)
{
    float L;                                /* metres */
    float v_man = 0.3, f_man = 0.30;        /* m/min */
    float v_rob = 0.5, f_rob = 0.80;
    float t_man, t_rob;

    printf("Enter weld length (m): ");
    scanf("%f", &L);

    t_man = L / (v_man * f_man) / 60.0;    /* h */
    t_rob = L / (v_rob * f_rob) / 60.0;

    printf("Manual : %.1f h\n", t_man);
    printf("Robot   : %.1f h\n", t_rob);
    printf("Time saved = %.1f %%\n",
           100.0 * (1.0 - t_rob / t_man));
    return 0;
}
```

Program Output

Enter weld length (m):
200
Manual : 37.0 h
Robot : 8.3 h
Time saved = 77.5 %

Check by hand ($L = 200$ m):

$$\text{Manual: } \frac{200}{0.3 \times 0.3} = 2222$$

min = 37.0 h

$$\text{Robot: } \frac{200}{0.5 \times 0.8} = 500$$

min = 8.3 h

Most of the gain comes from the higher **arc-on time**, not the higher speed.

Surface Preparation and Painting in Shipbuilding

Why it matters

Coating protects the hull and tanks from corrosion for 15–25 years. Rules such as IMO **PSPC** (ballast tanks) demand surface cleanliness (e.g. Sa 2½), profile and dry-film thickness (DFT) — consistently.

Why automate it

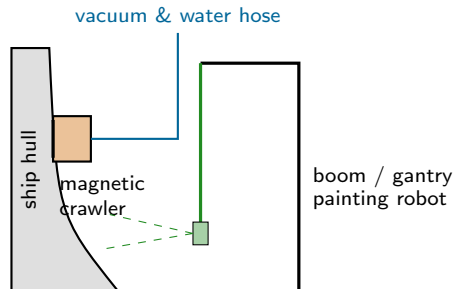
- Abrasive dust, silica, noise, solvent fumes — serious **health hazards**.
- Huge areas (tens of thousands of m² per ship).
- Uniform DFT is hard to achieve by hand.

Stage	Automated equipment
Plate / profile	shot-blast & priming line (fixed automation)
Blocks	blasting & painting robots in enclosed halls
Hull in dock	boom / gantry robots, magnetic crawlers
Repair & maintenance	UHP water-jetting crawlers, hull-cleaning ROVs

Blasting and Painting Robots

- **Magnetic crawlers** climb the steel hull and carry a blasting or UHP water-jet head; debris is vacuumed back (closed loop, no dust cloud).
- **Boom / gantry painting robots** spray the hull side in the dock with constant stand-off distance and speed.
- **Block painting robots** in halls with ventilation and temperature control.
- Sensors measure stand-off, overlap and wet-film thickness; data logged for inspectors.

Robots keep the gun at a constant distance, angle and speed — the three things that decide film thickness.



Painting: Worked Example on Paint Consumption

Paint volume needed

$$V_{\text{paint}} = \frac{A \times \text{DFT}}{\text{VS} \times \text{TE}}$$

A = area, DFT = dry-film thickness, VS = volume solids of the paint, TE = transfer efficiency (fraction reaching the surface).

Problem: a hull area of 5000 m^2 gets one coat of $\text{DFT } 250 \text{ }\mu\text{m}$, paint $\text{VS} = 60\%$. Manual spraying reaches $\text{TE} = 60\%$; a robot reaches $\text{TE} = 85\%$. How much paint is saved?

$$A \cdot \text{DFT} = 5000 \times 250 \times 10^{-6} = 1.25 \text{ m}^3$$

$$V_{\text{man}} = \frac{1.25}{0.6 \times 0.6} = 3.47 \text{ m}^3 \approx 3472 \text{ L}$$

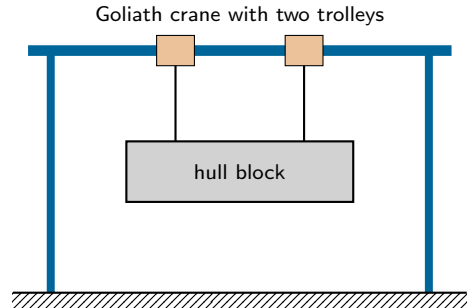
$$V_{\text{rob}} = \frac{1.25}{0.6 \times 0.85} = 2.45 \text{ m}^3 \approx 2451 \text{ L}$$

$$\text{Saving} \approx \boxed{1021 \text{ L (29\%)}}$$

Less overspray also means less solvent emission and cleaner air in the dock.

Heavy Lifting in the Shipyard

- **Goliath (gantry) cranes** over the building dock — lift blocks of several hundred to 1000+ tonnes.
- **Block turn-over** — two hooks rotate a block so it can be welded down-hand.
- **Self-propelled modular transporters (SPMT)** — computer-steered multi-axle trailers move blocks between shops.
- **Automated plate stock-yards** — magnet cranes pick plates by ID.



Automation: synchronised hoisting, anti-sway control, load monitoring, laser positioning.

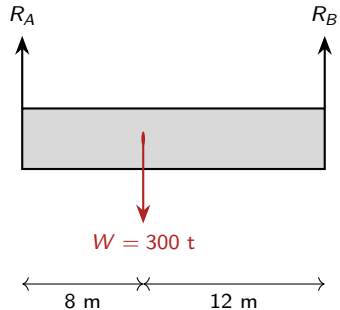
Heavy Lifting: Worked Examples

Tandem (two-hook) lift

A 300 t block hangs from hooks A and B, 20 m apart. Its centre of gravity is 8 m from A. Find the hook loads.

$$R_A = W \frac{12}{20} = 300 \times 0.6 = \boxed{180 \text{ t}}$$
$$R_B = W \frac{8}{20} = 300 \times 0.4 = \boxed{120 \text{ t}}$$

The hook **nearer** the CG carries more; each hoist must stay within its rating.



SPMT axle load

600 t block, 24 axle lines:

$600/24 = \boxed{25 \text{ t/axle line}}$, levelled hydraulically by the transporter's computer.

Automation of Other Shipyard Tasks

Task	Automation	Benefit
Plate cutting & marking	CNC plasma, oxy-fuel, laser; ink-jet marking	accurate parts, nested to save steel
Plate bending	CNC press, automated line heating	less skill-dependent forming
Pipe fabrication	robotic cutting, bending, orbital welding	high-quality spools, labels from CAD
Inspection	automated UT, 3-D laser scanning, drones in tanks	fast, safe, digital records
Logistics	RFID tracking, AGVs, automated warehouses	right part at the right time
Worker support	exoskeletons, AR glasses, cobots	less fatigue & injury

Beyond the yard

Automation also runs the ship itself: unmanned machinery spaces, integrated bridge systems, dynamic positioning and, increasingly, **autonomous vessels**.

Accuracy Control with 3-D Scanning and Drones

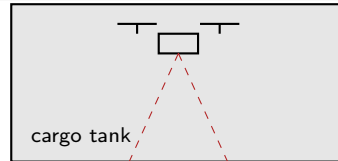
Accuracy control

Blocks must fit at erection within a few millimetres. A 3-D laser scanner or total station measures the finished block; software compares it with the CAD model and predicts the fit **before** the block is lifted.

- Less cutting and re-fitting in the dock.
- Shrinkage data fed back to design (added margins).

Drones & crawlers for inspection

Drones with cameras and ultrasonic probes inspect tanks, cargo holds and hull areas that would otherwise need scaffolding — faster and safer surveys.



camera / UT

Summary & What Comes Next

Concept

- Manufacturing concepts & production metrics.
- Reasons for automating; USA principle.
- Types of production & plant layouts in a shipyard.
- Types of automation, strategies, migration.
- Levels of automation in shipbuilding.
- Digital shipyards, product model, digital twin.
- Robots in welding, blasting, painting, lifting & other tasks.

Practice makes perfect

- Re-work the MLT, welding-time, paint and lifting examples with new data.
- Run the welding-time C program for other weld lengths and arc-on factors.
- List the automation you would add first in a small shipyard, and why.

Coming up

Marine automation on board: engine-room automation, integrated bridge systems and autonomous ships.

Thank You

Questions?